

Does Fisher–Rao Birth–Death Help Adaptive Biasing Force?

A Three-Case Study from a Model Potential to a Condensed-Phase Dimer

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Abstract

We study whether augmenting the Adaptive Biasing Force (ABF) method with a Fisher–Rao (FR) birth–death correction improves the computation of a reaction-coordinate free energy, and, if so, how the correction must be constructed. We test this on *three* systems of increasing difficulty, each with an essentially exact reference: a two-dimensional double-well *metastability* model (quadrature-exact reference); a two-dimensional *entropic-bottleneck* model with a narrow transverse channel, for which both the reference free energy and the conditional law $Y \mid X = x \sim \mathcal{N}(0, 1/(\beta\omega(x)^2))$ are analytic; and a many-body condensed-phase *Weeks–Chandler–Andersen (WCA) dimer* in a dense solvent (thermodynamic-integration reference, GPU). In each case the FR step reallocates walkers along the reaction coordinate by killing replicas in over-represented regions and cloning whole configurations from under-represented regions, nudging the empirical marginal $\hat{p}_t$ toward a target q_t . We compare ABF only against ABF+FR with an *estimated*, *uniform*, or *oracle* target, on matched seeds. The central finding is that **FR helps ABF in proportion to how sample-starved ABF is**: a modest convergence accelerator — and occasionally mildly harmful — where plain ABF already samples well (metastability, +11.5% integrated error; hot bottleneck, −26.1%), but a large and reliable win where ABF is starved (cold bottleneck, +51.3%, 20/20 seeds; WCA dimer, +55.0%, 10/10 seeds, the relative gain growing with simulation budget and the largest absolute error removed in the densest solvent). The dominant mechanism is *balanced-resampling variance reduction*, not free-energy-shape steering: in the starved regimes a uniform target nearly matches the estimated one and an oracle target (which is allowed to use the reference F_{ref} , and is never a deployable method) is no better — or, in the entropic bottleneck, *worse* than ABF. The one failure mode is diversity collapse under too-aggressive birth–death (the WCA dimer’s gain reverses once `fr_rate` is pushed well past its sweet spot), bounded by how fast the system’s orthogonal degrees of freedom re-equilibrate cloned replicas. Across all successful settings FR preserves the conditional law $Y \mid X = x$ on which ABF depends — confirmed directly against the analytic conditional of the bottleneck model.

1 Introduction

1.1 Free-energy computation along a reaction coordinate

A central task in computational statistical mechanics is to compute the free energy associated with a low-dimensional *reaction coordinate* of a high-dimensional system in thermal equilibrium [8]. For

most of the development we take a configuration to be a point $q = (x, y) \in \mathbb{R}^{21}$ drawn from the canonical (Boltzmann–Gibbs) density

$$\pi(x, y) \propto e^{-\beta V(x, y)}, \quad (1)$$

where V is the potential energy and $\beta = 1/(k_B T)$ is the inverse temperature. Fixing the reaction coordinate to the first coordinate, $\xi(x, y) = x$, the free energy $F(x)$ and the (local) mean force $F'(x)$ along ξ are the objects of interest. The difficulty is metastability: when V has well-separated basins along x , unbiased Langevin dynamics crosses the barrier rarely, so a direct estimate of F converges intolerably slowly.

1.2 Adaptive Biasing Force

The Adaptive Biasing Force (ABF) method [3, 4, 6] removes this metastability adaptively. It maintains a running estimate $\widehat{F}'_t(x)$ of the mean force and applies the antiderivative $B_t(x) = \int^x \widehat{F}'_t$ as a bias that depends only on x . As $B_t \rightarrow F$, the effective dynamics along x becomes diffusive and the reaction coordinate is explored freely, while — crucially — the bias does not change the conditional law of the orthogonal degrees of freedom given x . ABF is mathematically well understood, with long-time convergence guarantees [7], and is a natural baseline against which to measure any proposed enhancement.

1.3 Why add a Fisher–Rao correction?

ABF accelerates exploration through the *force*, but it does not directly control *where the particles are*. Early in a simulation the estimated bias is poor, so the empirical reaction-coordinate marginal $\widehat{p}_t(x)$ can remain concentrated in the initial basin, starving other values of x of the samples ABF needs to estimate $F'(x)$ there. A Fisher–Rao (FR) birth–death correction [9] addresses exactly this: interpreting the evolution of $\widehat{p}_t$ as a gradient flow of relative entropy in the Fisher–Rao geometry [1], one reallocates probability mass by *killing* particles in over-represented regions and *cloning* particles in under-represented regions, driving $\widehat{p}_t$ toward a chosen target marginal $q_t(x)$ without integrating any additional dynamics. Coupling such a correction to ABF is appealing because the two act on complementary objects (force versus density), but it is not obvious that it helps: a birth–death step resamples *whole* configurations (x, y) , and if it distorts the conditional law $Y \mid X = x$ it will corrupt precisely the quantity ABF estimates.

1.4 Central question and contributions

This report asks three questions across *three* controlled systems of increasing difficulty, each with an essentially exact reference free energy:

1. **Does FR help ABF?** (Q1)
2. **How should FR be added so that it helps?** (Q2–Q3)
3. **What are the limitations, failure modes, and future directions?** (Q4)

¹We write a configuration as $q = (x, y)$ in the two-dimensional cases; the many-body case of Section 6 uses $q \in \mathbb{R}^{2n}$ with a scalar collective variable $z(q)$ in place of x (Remark 1). The symbol $q_t(\cdot)$, always carrying a time subscript and a scalar argument, denotes instead the Fisher–Rao *target marginal* of Section 3; the two are never used in the same role.

The three systems are (i) a two-dimensional double-well *metastability* model with a quadrature-exact reference (Section 4); (ii) a two-dimensional *entropic-bottleneck* model with a narrow transverse channel, for which both the reference free energy *and* the conditional law $Y \mid X = x$ are analytic (Section 5); and (iii) a many-body *Weeks–Chandler–Andersen (WCA) dimer* in a dense solvent with a thermodynamic-integration reference (Section 6). Together they span a range of *how sample-starved ABF is*: from a regime where plain ABF is already a strong baseline to regimes where it is badly starved on the far side of a barrier.

Our contributions are: (i) a precise statement of the coupled ABF–FR algorithm in consistent notation, applicable to both two-dimensional and many-body systems (Sections 2 and 3); (ii) three matched-seed production studies that quantify the effect of FR and of its target marginal on each system (Sections 4 to 6); and (iii) a cross-case synthesis (Section 7) that ties the three results into a single picture. A recurring methodological point is the careful interpretation of an *oracle* target that uses the reference free energy: it is a diagnostic control that bounds the achievable benefit, never a deployable method. The central finding, developed in Section 7, is that *FR helps ABF in proportion to how starved ABF is*: a modest convergence accelerator (or even mildly harmful) when ABF already samples well, but a large and reliable variance-reduction gain when ABF is starved — and that in the starved regimes the precise target *shape* matters surprisingly little.

2 Background: ABF for a reaction-coordinate free energy

2.1 Free energy and the unbiased marginal

With reaction coordinate $\xi(x, y) = x$, the unbiased reaction-coordinate marginal of the canonical density (1) is

$$p_{\text{ref}}(x) = \frac{\int e^{-\beta V(x, y)} \, dy}{\iint e^{-\beta V(x, y)} \, dx \, dy}, \quad (2)$$

and the free energy along ξ is, up to an additive constant,

$$F_{\text{ref}}(x) = -\frac{1}{\beta} \log p_{\text{ref}}(x) + C. \quad (3)$$

Because F is defined only up to the constant C , every free-energy profile in this report — both estimates and the reference — is *centred* (its mean over the interior evaluation window is subtracted) before any L^2 error is formed; otherwise a meaningless constant offset would dominate the comparison. The reference quantities are computed by a one-dimensional y -quadrature on a fine grid with a log-sum-exp shift for numerical stability, which is exact up to discretisation and provides the ground truth $F_{\text{ref}}, F'_{\text{ref}}, p_{\text{ref}}$.

2.2 The mean force is the object ABF estimates

Differentiating (3) and using $\partial_x p_{\text{ref}}$ gives the *mean force*

$$F'_{\text{ref}}(x) = \frac{\int \partial_x V(x, y) e^{-\beta V(x, y)} \, dy}{\int e^{-\beta V(x, y)} \, dy} = \mathbb{E}_{\pi}[\partial_x V(X, Y) \mid X = x]. \quad (4)$$

Equation (4) is the cornerstone of ABF: the gradient of the free energy equals the *conditional expectation of the instantaneous force* $\partial_x V$ given the reaction coordinate. It is a local, sample-based quantity — no integral of V is required — so it can be estimated on the fly from the particles currently at $X \approx x$, and the free energy is then recovered by quadrature, $\hat{F}(x) = \int^x \hat{F}'$.

2.3 Biased dynamics

ABF simulates N overdamped Langevin replicas that share one running mean-force estimate $\hat{F}'_t(x) \approx F'_{\text{ref}}(x)$ with bias potential $B_t(x) = \int^x \hat{F}'_t$. Each replica (X_t, Y_t) evolves as

$$\begin{aligned} dX_t &= [-\partial_x V(X_t, Y_t) + \hat{F}'_t(X_t)] dt + \sqrt{2/\beta} dW_t^x, \\ dY_t &= -\partial_y V(X_t, Y_t) dt + \sqrt{2/\beta} dW_t^y, \end{aligned} \quad (5)$$

with independent Brownian motions W^x, W^y . The added force $+\hat{F}'_t(X_t)$ acts *only* in the x -direction; the y -dynamics is unbiased. When $\hat{F}'_t \rightarrow F'_{\text{ref}}$ the net drift along x becomes $-\partial_x(V - F)$, whose x -marginal equilibrium is flat, so the reaction coordinate diffuses freely across the barrier.

2.4 Why the bias does not spoil the mean-force estimate

A bias that depends only on x leaves the conditional law of Y given $X = x$ invariant. This is what makes ABF self-consistent.

Proposition 1 (Conditional invariance under an x -only bias). *Let ν be the canonical measure $\nu \propto e^{-\beta V(x,y)}$ and let $\nu_B \propto e^{-\beta[V(x,y)-B(x)]}$ be the measure biased by any function B of x alone. Then for every x the conditionals coincide, $\nu_B(y | X = x) = \nu(y | X = x)$, and consequently*

$$\mathbb{E}_{\nu_B}[\partial_x V(X, Y) | X = x] = \mathbb{E}_{\nu}[\partial_x V(X, Y) | X = x] = F'_{\text{ref}}(x). \quad (6)$$

Proof. At fixed $X = x$ the bias factor $e^{\beta B(x)}$ is a constant in y and cancels between numerator and normaliser of the conditional density:

$$\nu_B(y | X = x) = \frac{e^{-\beta[V(x,y)-B(x)]}}{\int e^{-\beta[V(x,y')-B(x)]} dy'} = \frac{e^{-\beta V(x,y)}}{\int e^{-\beta V(x,y')} dy'} = \nu(y | X = x).$$

Taking the conditional expectation of $\partial_x V$ under this common conditional gives (6), the last equality being (4). $\square$

Proposition 1 is the reason ABF works: even though the biased ensemble (5) has a different x -marginal than the canonical one, the conditional samples $Y | X = x$ remain canonical, so the empirical conditional average of $\partial_x V$ is an unbiased target for $F'_{\text{ref}}(x)$. Any modification of ABF must respect this property; in particular, a birth–death correction that resamples whole configurations (x, y) is only admissible if it too leaves $Y | X = x$ (approximately) intact — a hypothesis we test directly in each of the three case studies (Sections 4 to 6).

Remark 1 (Generalisation to many-body systems). Two of our three systems are two-dimensional with $\xi = x$, but the WCA dimer of Section 6 is a many-body system: a configuration is the full vector $q \in \mathbb{R}^{2n}$ of all particle coordinates, and the reaction coordinate is a scalar *collective variable* $z(q)$ (a scaled bond length) rather than a single Cartesian component. Proposition 1 holds verbatim under the substitution $x \mapsto z(q)$ and $y \mapsto$ the orthogonal complement of z on the constant- z manifold (here the solvent coordinates and the transverse dimer modes): a bias $B(z)$ that depends on the collective variable alone is constant on each level set of z , so it cancels in the

conditional density and leaves the law of all orthogonal degrees of freedom invariant. Consequently the mean force along z — which for a nonlinear $z(q)$ also carries a geometric term (Section 6) — is still estimated without bias. The only requirement the FR step must meet is the same one as in two dimensions: a clone must copy the *entire* configuration $q \in \mathbb{R}^{2n}$, so that the orthogonal coordinates travel with z and the conditional law is preserved.

2.5 The ABF mean-force estimator

In the kernel form used by the reference implementation, $\hat{F}'$ is a Nadaraya–Watson running average of the instantaneous force,

$$\hat{F}'_n(x) = \frac{\sum_{k \leq n} \sum_i K_h(x - X_k^i) \partial_x V(X_k^i, Y_k^i)}{\sum_{k \leq n} \sum_i K_h(x - X_k^i)}, \quad (7)$$

with a Gaussian kernel K_h of bandwidth h , accumulated over all replicas i and past steps k ; the free-energy estimate is its pinned trapezoidal antiderivative, $\hat{F}_n(x) = \int_0^x \hat{F}'_n$, and we use $B_t \equiv \hat{F}$ as the bias. The GPU backend used for the production study replaces the $O(n_{\text{grid}} N)$ kernel sums by an $O(N + n_{\text{grid}})$ *binned-smoothed* estimator: particles are scattered onto the x -grid and the count and force histograms are Gaussian-smoothed, so that $\hat{F}'(x) = \text{smooth}(\text{force})/\text{smooth}(\text{count})$ is a Riemann-sum approximation of (7) that converges to it as the grid spacing vanishes. The two estimators are validated to agree to well under 1% on static tests (Section 4.1).

3 The Fisher–Rao marginal correction

The Fisher–Rao (FR) correction is a birth–death / particle-reallocation step that acts on the *reaction-coordinate marginal* of the ensemble. It is interleaved with the ABF dynamics (5) and is designed to drive the empirical marginal $\hat{p}_t(x)$ toward a target marginal $q_t(x)$, without adding any drift to the (x, y) dynamics.

3.1 Empirical and target marginals

At step n , after the Langevin proposal of all N replicas, the empirical reaction-coordinate marginal is estimated by a (reflected, edge-corrected) kernel density / binned-smoothed estimate of bandwidth η ,

$$\hat{p}_n(x) = \frac{1}{N} \sum_{i=1}^N K_\eta(x - X_n^i), \quad (8)$$

renormalised on the x -grid. The target $q_n(x)$ is one of three choices (Section 3.6); all are nonnegative and normalised, $\int q_n = 1$.

3.2 Fisher–Rao score

For each particle the FR *score* is the pointwise log-ratio of the empirical to the target marginal, with its ensemble mean removed:

$$S_i = \log \frac{\hat{p}_n(X_n^i) + \varepsilon}{q_n(X_n^i) + \varepsilon} - \underbrace{\int \hat{p}_n(z) \log \frac{\hat{p}_n(z)}{q_n(z)} dz}_{= \text{KL}(\hat{p}_n \parallel q_n)} \quad (9)$$

where $\varepsilon > 0$ is a small regulariser and the integral baseline equals the Kullback–Leibler divergence $\text{KL}(\hat{p}_n \| q_n)$ (a grid quadrature; it coincides with the Monte-Carlo baseline $\frac{1}{N} \sum_k \log[\hat{p}_n(X^k)/q_n(X^k)]$ up to discretisation). Subtracting the baseline makes the score mean-zero, so the birth–death step below conserves the expected particle number. The score is clipped to $[-S_{\max}, S_{\max}]$ for stability. Its interpretation is direct:

- $S_i > 0$: X^i lies where $\hat{p}_n > q_n$, an *over-represented* region (too many particles);
- $S_i < 0$: X^i lies where $\hat{p}_n < q_n$, an *under-represented* region (too few particles).

3.3 Birth–death resampling

The FR step realises the Fisher–Rao gradient flow of $\text{KL}(\hat{p} \| q)$ as a fixed- N birth–death process. Over a sub-step of length Δt and at effective rate γ_n^{eff} (Section 3.5), each *over-represented* particle ($S_i > 0$) is killed with probability

$$\mathbb{P}(i \text{ dies}) = 1 - \exp(-\gamma_n^{\text{eff}} S_i \Delta t), \quad (10)$$

and every killed slot is overwritten by a *clone* of an *under-represented* particle, sampled from the pool $\{i : S_i < 0\}$ with weights $\propto |S_i|$. Cloning copies the *whole* configuration (X^i, Y^i) — the score depends only on x , but the y -coordinate must travel with its x -coordinate, or Proposition 1 would be violated. The step is made deliberately gentle by three safeguards: (i) a burn-in and soft ramp on the rate; (ii) score clipping to $[-S_{\max}, S_{\max}]$; and (iii) a hard cap $f_{\max}$ on the fraction of particles replaced per event (`max_event_fraction`), enforced by randomly sub-sampling deaths when the cap is exceeded. The numerical values of $S_{\max}$, $f_{\max}$ and the FR stride `fr_every` are case-specific and are listed in each case’s design table. Net effect: gently replace particles in crowded regions of x by copies from starved regions, leaving N fixed.

3.4 Notation across cases

The same FR mechanism is applied to all three systems, but two parameters carry different names in the analytic two-dimensional cases (Sections 4 and 5) and in the many-body WCA study (Section 6), for consistency with each study’s code. The birth–death *rate* is written γ in the analytic cases and `fr_rate` in WCA; they play identical roles in (10). The *onset* of FR is parameterised as a burn-in *fraction* b in the analytic cases ((11), $n_{\text{burnin}} = \lfloor b n_{\text{steps}} \rfloor$) and as an absolute step count `fr_start_steps` in WCA; the two are related by $b \approx \text{fr_start_steps}/n_{\text{steps}}$. Where a case uses a collective variable $z(q)$ rather than x , read $x \mapsto z$ throughout this section (Remark 1).

3.5 Rate schedule

The FR rate is zero during an initial burn-in of $n_{\text{burnin}} = \lfloor b n_{\text{steps}} \rfloor$ steps (burn-in fraction b), then ramps softly to its maximum γ ,

$$\gamma_n^{\text{eff}} = \gamma \left(1 - e^{-(n - n_{\text{burnin}})/n_{\text{ramp}}} \right), \quad n > n_{\text{burnin}}, \quad (11)$$

and the birth–death step is applied every `fr_every` steps. The burn-in lets ABF build a usable mean-force estimate (hence a usable target) before any particles are moved; the ramp avoids a sudden shock to the ensemble.

3.6 Target marginals: estimated, uniform, oracle

The three targets differ only in which density q_n the score (9) is taken against. Recall $B_t = \hat{F}_n$ is the current ABF bias.

Estimated target (the practical method). A target free energy $\hat{F}_n^{\text{target}}$ is maintained independently of the bias by an exponential moving average of the ABF estimate, $\hat{F}_{n+1}^{\text{target}} = (1-r)\hat{F}_n^{\text{target}} + r\hat{F}_{n+1}$, and the target marginal is the *biased* marginal that this target free energy would induce under the current bias,

$$q_n^{\text{est}}(x) \propto \exp\left[-\beta(\hat{F}_n^{\text{target}}(x) - B_t(x))\right]. \quad (12)$$

This uses only quantities available at run time and is the method whose practical value we assess.

Uniform target (a naive ablation).

$$q_n^{\text{unif}}(x) = \text{Unif}(x) \quad \text{on the } x\text{-grid}. \quad (13)$$

This tests whether simply *flattening* the reaction-coordinate marginal is enough. It is a target ablation, not the intended method; it ignores the fact that, under a bias $B_t \neq F$, the canonical biased marginal is not uniform.

Oracle target (a diagnostic control only).

$$q_n^{\text{oracle}}(x) \propto \exp\left[-\beta(F_{\text{ref}}(x) - B_t(x))\right]. \quad (14)$$

Remark 2 (The oracle target is not an algorithm). Equation (14) replaces the estimated target free energy by the *reference* free energy F_{ref} — which is exactly the unknown quantity the whole procedure is trying to compute. The oracle target therefore *cannot* be evaluated in any real free-energy calculation. We include it solely as a *diagnostic / positive / ideal-information control*: it measures how much the FR mechanism *could* help if the target marginal direction were specified perfectly. Throughout this report the oracle is read only in that way; we never claim that “oracle FR is the best method” or that “one should use the oracle target”. The practical comparison is always

$$\text{ABF only} \quad \text{versus} \quad \text{ABF} + \text{estimated-target FR},$$

and the estimated-versus-oracle gap is interpreted as the headroom available from improving the online target estimate.

3.7 The coupled ABF–FR step

One outer iteration, given the state $\{(X_n^i, Y_n^i)\}$ and accumulators, is:

1. update the ABF accumulators from the current particles and recompute $\hat{F}_n', \hat{F}_n = B_t$ (Section 2.5);
2. propose the biased Langevin move (5) for all replicas;
3. if FR is active at this step (past burn-in, on the `fr_every` schedule): form $\hat{p}_n$ (8) and the target q_n — one of (12), (13), (14) — compute the scores (9), and apply the birth–death step (10);

4. record diagnostics at the evaluation cadence.

The ABF estimator sees the post-resampling ensemble at the next step, which is the route by which improved x -coverage feeds back into a better mean-force estimate. The remainder of the report quantifies whether, and how, this feedback is beneficial.

4 Case I: a two-dimensional metastability model

Our first system is a two-dimensional double well: the regime in which ABF is already a strong baseline, so it sets the lower end of the difficulty spectrum.

4.1 Model potential and reference

We use a two-dimensional model potential with reaction coordinate $\xi(x, y) = x$,

$$V(x, y) = 3e^{-x^2} \left(e^{-(y-1/3)^2} - e^{-(y-5/3)^2} \right) - 5e^{-y^2} \left(e^{-(x-1)^2} + e^{-(x+1)^2} \right) + 0.2x^4 + 0.2(y-1/3)^4. \quad (15)$$

As shown in Figure 1, V has two metastable basins near $x = \pm 1$ separated by a barrier near $x = 0$ (with an upper sub-basin near $(0, 5/3)$), so unbiased dynamics is metastable in x . We fix $\beta = 4$. The reference $F_{\text{ref}}, F'_{\text{ref}}, p_{\text{ref}}$ are obtained by the y -quadrature of (2)–(4) on a fine grid and are treated as ground truth.

4.2 Methods, metrics, and study design

Four methods are compared, identical except for the FR target: **ABF only** (no birth–death; baseline); **ABF+FR estimated** (12) (the practical method); **ABF+FR uniform** (13) (naive marginal-flattening ablation); and **ABF+FR oracle** (14) (diagnostic control, Remark 2). All four share the same potential, integrator, ABF estimator, seeds and initial conditions, so differences are attributable to the FR step and its target alone (matched-seed comparison).

4.2.1 Metrics

All profile errors are interior, domain-RMS L^2 distances on the window $x \in [-2.5, 2.5]$ (the reflecting walls and the x^4 confinement make the edges uninformative), with free-energy profiles centred as in Section 2.

Primary (practical) metrics.

$$\left\| \widehat{F}_T - F_{\text{ref}} \right\|_{L^2}, \quad \left\| \widehat{F}'_T - F'_{\text{ref}} \right\|_{L^2}, \quad \int_0^T \left\| \widehat{F}_t - F_{\text{ref}} \right\|_{L^2} dt. \quad (16)$$

The first two are *final-budget accuracy* at the end of the run; the *integrated* error rewards *fast* convergence (anytime performance), since a method that reaches a good profile earlier accumulates less error over $[0, T]$. These two notions need not be optimised by the same hyperparameters, a point we treat carefully in Section 4.3.

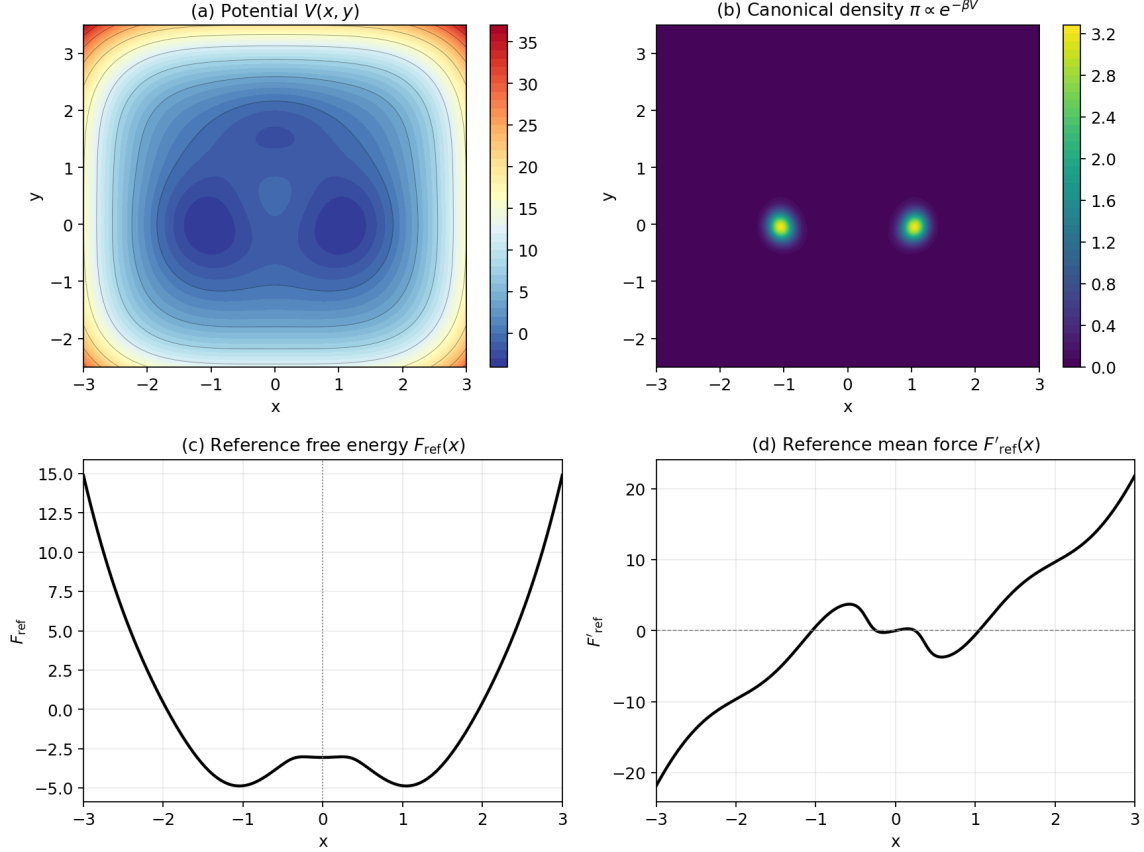


Figure 1: Problem geometry. (a) The potential $V(x, y)$ (15) with two basins near $x = \pm 1$. (b) The canonical density $\pi \propto e^{-\beta V}$ at $\beta = 4$, concentrated in the two basins. (c) The reference free energy $F_{\text{ref}}(x)$ (a double well in x) and (d) the reference mean force $F'_{\text{ref}}(x)$, both by y -quadrature.

Diagnostic metrics. Reaction-coordinate marginal flatness ($\|\hat{p} - \text{Unif}\|$ and $\|\hat{p} - q\|$); cumulative barrier crossings at $x = 0$; region fractions (left basin / barrier / right basin); the FR event fraction (particles replaced per FR application) and the FR score dispersion $\text{std}_i S_i$; conditional $Y | X = x$ errors (Section 4.2.1); and ancestor/diversity counts. We stress that *marginal flatness is a diagnostic, not the objective*: the goal is free-energy/mean-force accuracy (16). A method can flatten $\hat{p}$ and yet fail to improve $\hat{F}$ if it does so by corrupting the conditional samples.

Conditional $Y | X$ diagnostics. At a set of x -bin centres we compare the empirical conditional $\hat{p}(y | x \in I_j)$ (the y -histogram of particles whose x falls in a narrow window) to the reference conditional $p_{\text{ref}}(y | x) \propto e^{-\beta V(x, y)}$, via an L^2 distance and a KL divergence. By Proposition 1 a benign FR step should leave these close to the ABF-only values.

4.2.2 GPU implementation and study size

The production runs use a batched PyTorch backend that simulates many runs at once and reuses the reference metrics/diagnostics code verbatim. For speed it uses the binned-smoothed ABF/FR estimator of Section 2.5 rather than the exact kernel sums (7), and a fully vectorised fixed- N birth-death; the two backends agree to within the stated tolerance ($< 1\%$ in practice) on the validation suite. We flag in Section 8 that this binned estimator is a deliberate, but not mathematically

Table 1: Production study design (read from the result CSVs and configs/two_dim_xi_x_production_gpu.yaml).

Quantity	Value
Inverse temperature β	4.0
Time step Δt	0.002
Steps per run	100,000
Particles per run N	1000
Seeds	5 (0, 1, 2, 3, 4)
Methods	4 (ABF; FR estimated / uniform / oracle)
FR rate γ	0.01, 0.02, 0.05, 0.1
FR bandwidth η	0.075, 0.1, 0.15
FR burn-in fraction	0, 0.2, 0.4
FR stride <code>fr_every</code>	5
Total configurations	109
Total runs	545

Table 2: Hyperparameters of the selected configurations (integrated- $L^2(F)$ rule). n_{FR} is the Fisher–Rao stride `fr_every`.

Method	target	γ	η	burn-in	n_{FR}	seeds
ABF only	none	0	0.075	0	1	5
FR estimated	estimated	0.1	0.075	0	5	5
FR uniform	uniform	0.05	0.15	0.2	5	5
FR oracle*	oracle	0.1	0.075	0	5	5

identical, surrogate for the kernel notebook implementation. The FR methods sweep the rate γ , the marginal bandwidth η , and the burn-in fraction b at a fixed FR stride; the resulting study (Table 1) comprises 545 runs over 109 configurations and 5 seeds, at 1000 particles and 100,000 steps each, with *no* NaNs in the primary metrics. The selected configurations are listed in Table 2.

4.3 Results

Table 3 reports the four methods, each at its best configuration under the *integrated- $L^2(F)$* rule (convergence speed; see Section 4.3.2). Reading the practical comparison (ABF only versus ABF+FR estimated): the estimated-target FR reduces the median integrated free-energy error from 15.91 to 14.08, a 11.5% reduction, while the *final* free-energy error improves only slightly (from 0.0137 to 0.0126). The birth–death is extremely gentle: the median per-event replacement fraction is 1.6×10^{-5} , of order one particle in 10^5 per FR application. Figure 2 shows the same story over time: the FR curves detach below the ABF-only curve in the early and middle phase of the $L^2(F)$ panel.

4.3.1 Final profiles and target-type ablation

Figure 3 overlays the final mean-force, free-energy and reaction-coordinate-marginal profiles. All practical methods recover F'_{ref} and F_{ref} well; the visible differences between ABF only and ABF+FR estimated at the final budget are small, consistent with the modest final-error improvements. The x -marginal (panel c) is somewhat flatter under FR, as intended, but marginal flatness is a diagnostic, not the goal.

Table 3: Main comparison, configurations selected by smallest median *integrated* $L^2(F)$ (convergence speed). Lower is better except P (beats ABF).

Method	$\int_0^T \ \hat{F}_t - F_{\text{ref}}\ dt$	$\ \hat{F}_T - F_{\text{ref}}\ $	$\ \hat{F}'_T - F'_{\text{ref}}\ $	P (beats ABF)	FR event frac.
ABF only	15.91	0.0137	0.0718	–	0
FR estimated	14.08	0.0126	0.0771	0.60	1.6×10^{-5}
FR uniform	14.46	0.0125	0.0769	0.80	5.3×10^{-6}
FR oracle*	13.27	0.0113	0.0602	1.00	1.7×10^{-5}

* The oracle target uses the reference free energy F_{ref} (the unknown being computed); it is a diagnostic positive control, *not* a usable algorithm. Per-config medians over 5 seeds; one configuration per method, selected by the smallest median integrated $L^2(F)$.

Table 4: Supplement: configurations selected by smallest median *final* $L^2(F)$ (final-budget accuracy).

Method	$\int_0^T \ \hat{F}_t - F_{\text{ref}}\ dt$	$\ \hat{F}_T - F_{\text{ref}}\ $	$\ \hat{F}'_T - F'_{\text{ref}}\ $	P (beats ABF)	FR event frac.
ABF only	15.91	0.0137	0.0718	–	0
FR estimated	15.25	0.0120	0.0673	0.80	1.1×10^{-6}
FR uniform	14.97	0.0118	0.0680	0.60	7.3×10^{-6}
FR oracle*	13.27	0.0113	0.0602	1.00	1.7×10^{-5}

* Oracle is a diagnostic control (uses F_{ref}), not a deployable method. Configurations here are selected by smallest median *final* $L^2(F)$; cf. Table 3, which need not pick the same hyperparameters.

Figure 4 shows the distribution, over the 36 configurations of each target type, of the seed-median errors. For the *integrated* error all three FR target types have medians below the ABF-only baseline, but the spread is wide and some configurations are *worse* than ABF, so FR does not help unconditionally. For the *final* $L^2(F)$ the estimated and uniform medians sit essentially on the ABF baseline, and on the final *mean-force* error they lie slightly *above* it — only the oracle control is consistently below. Flattening the marginal with an imperfect target buys early speed but not final mean-force accuracy.

4.3.2 Two selection rules give different “best” hyperparameters

Because integrated and final errors measure different things, they need not be optimised by the same configuration. Selecting by *integrated* $L^2(F)$ (Table 3) favours fast convergence and selects an aggressive configuration for the estimated target ($\gamma = 0.1$, burn-in 0); selecting by *final* $L^2(F)$ (Table 4) favours final-budget accuracy and selects a gentle, delayed configuration ($\gamma = 0.01$, burn-in 0.4), which improves the final free-energy error by 12.5% and beats ABF on 80.0% of matched seeds. This dependence on the selection rule is itself a finding.

The *uniform* target is competitive here (integrated reduction 9.1%): under a near-converged bias $B_t \approx F$ the canonical biased marginal is already close to uniform, so “flatten $\hat{p}$ ” and “match the biased marginal” nearly coincide. The *oracle* diagnostic (Remark 2) achieves the lowest errors of all (integrated reduction 16.6%), but it requires F_{ref} and is unusable; the correct reading is comparative: the estimated target captures 11.5% of the reduction against the oracle’s 16.6%, so a substantial part of the mechanism’s potential is left unrealised by the online target estimate. In this easy regime, then, the oracle–estimated gap localises the practical bottleneck in the *target estimation* — a reading we revisit, and qualify, in Section 7.

Convergence of the selected (best integrated $L^2(F)$) configuration: median and IQR over 5 seeds

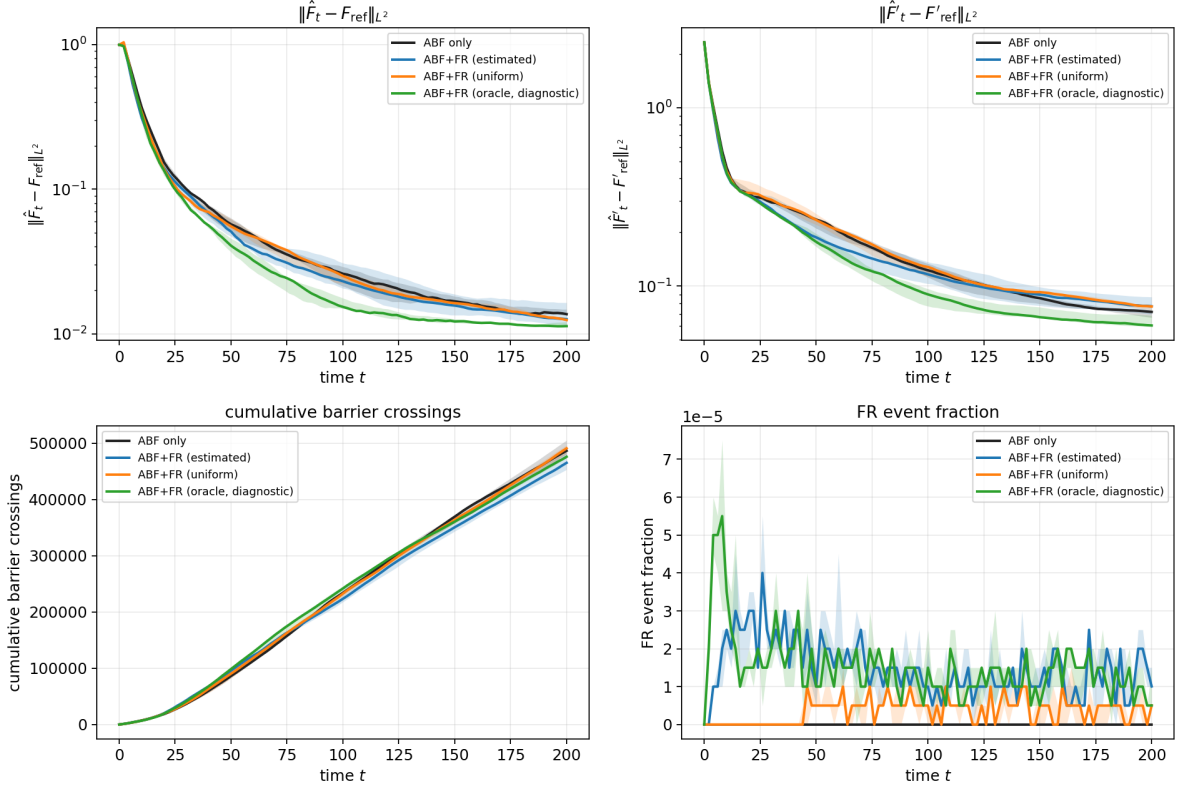


Figure 2: Convergence of the selected (integrated- $L^2(F)$ rule) configuration for each method: median (line) and inter-quartile range (band) over 5 seeds. The free-energy and mean-force errors (top, log scale) fall fastest for the oracle diagnostic and faster for the estimated/uniform FR than for ABF only in the early-to-middle phase; barrier crossings (bottom left) are essentially indistinguishable, and the FR event fraction (bottom right) is of order 10^{-5} throughout. The oracle curve is a diagnostic control (Remark 2).

4.4 Mechanism

The mechanism is a feedback loop between coverage and the mean-force estimate. ABF needs samples *across* the reaction coordinate: by (4), $F'_{\text{ref}}(x)$ is a conditional average that can only be estimated where particles visit x . Early in a run the bias is poor and the ensemble is unbalanced — Figure 5(b) shows the initial occupancy split (~ 0.5 in one basin) relaxing only gradually toward the balanced $1/3$ – $1/3$ – $1/3$ split. FR compares $\hat{p}_t$ with the target q_t and preferentially kills over-represented and clones under-represented particles (10), raising the useful coverage of x earlier than diffusion alone would. Provided the birth–death does not corrupt $Y \mid X = x$ (Proposition 1), better coverage feeds back into a better conditional mean-force estimate.

A notable feature of Figure 5(c) is that FR does *not* produce many more barrier crossings than ABF; the benefit comes from *reallocating* existing replicas toward starved values of x rather than from manufacturing rare transitions. Consistent with this, the event fractions are of order 10^{-5} : in the successful regimes FR is a *weak* correction.

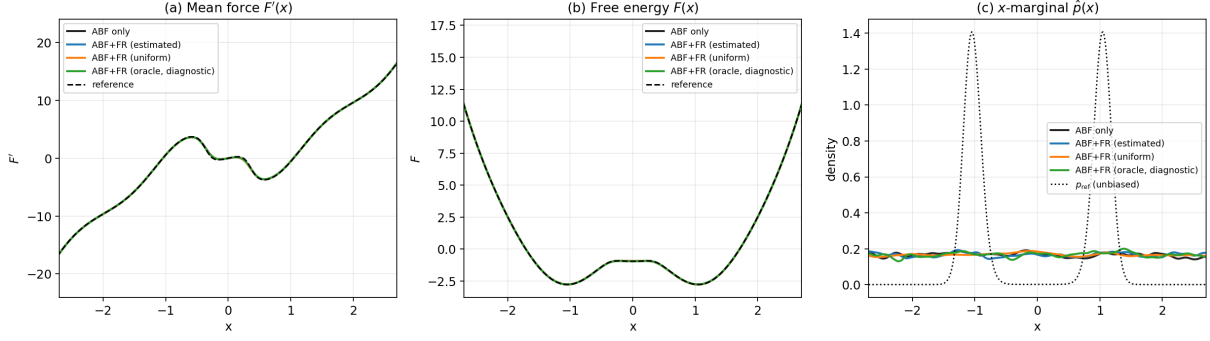


Figure 3: Final profiles (mean over seeds) for the selected configurations. (a) Mean force $\hat{F}'(x)$ and (b) centred free energy $\hat{F}(x)$ versus the reference (dashed); (c) reaction-coordinate marginal $\hat{p}(x)$, with the unbiased p_{ref} (dotted). Practical methods recover the reference well; FR mildly flattens $\hat{p}$.

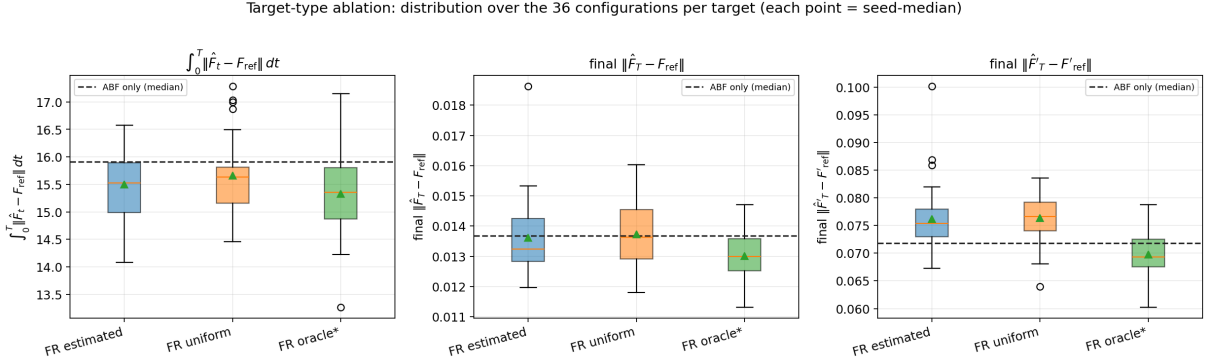


Figure 4: Target-type ablation: each box is the distribution over the 36 configurations of a target type (each point a seed-median); the dashed line is the ABF-only median. FR lowers the *integrated* error (left) but the practical targets barely change the *final* $L^2(F)$ (middle) and slightly raise the final mean-force error (right). Only the oracle diagnostic improves all three.

4.4.1 FR does not visibly corrupt $Y \mid X = x$

Figure 6 compares the conditional fidelity of all methods at the final time. The FR variants are *not* worse than ABF only; near the barrier ($x \in \{-0.5, 0, 0.5\}$), where ABF-only under-samples, they are in fact slightly *better* on both the L^2 and KL conditional metrics — direct evidence that the FR mechanism improves x -coverage without corrupting the orthogonal sampling, the benign behaviour Proposition 1 requires.

Remark 3 (Conditional diagnostics are not configuration-resolved). The metastability conditional-diagnostics output records only (method, target type, seed) per row, not the FR hyperparameters. It supports the *aggregate* statement that successful FR does not visibly corrupt $Y \mid X = x$, but cannot isolate a specific “good” versus “bad” configuration’s effect. The entropic-bottleneck case (Section 5) closes this gap with an *analytic* conditional law.

4.4.2 What determines a good FR?

Figure 7 (estimated target) shows the integrated error decreasing as γ grows — a stronger correction converges faster, which is why the integrated rule selects the aggressive $\gamma = 0.1$ — but the same

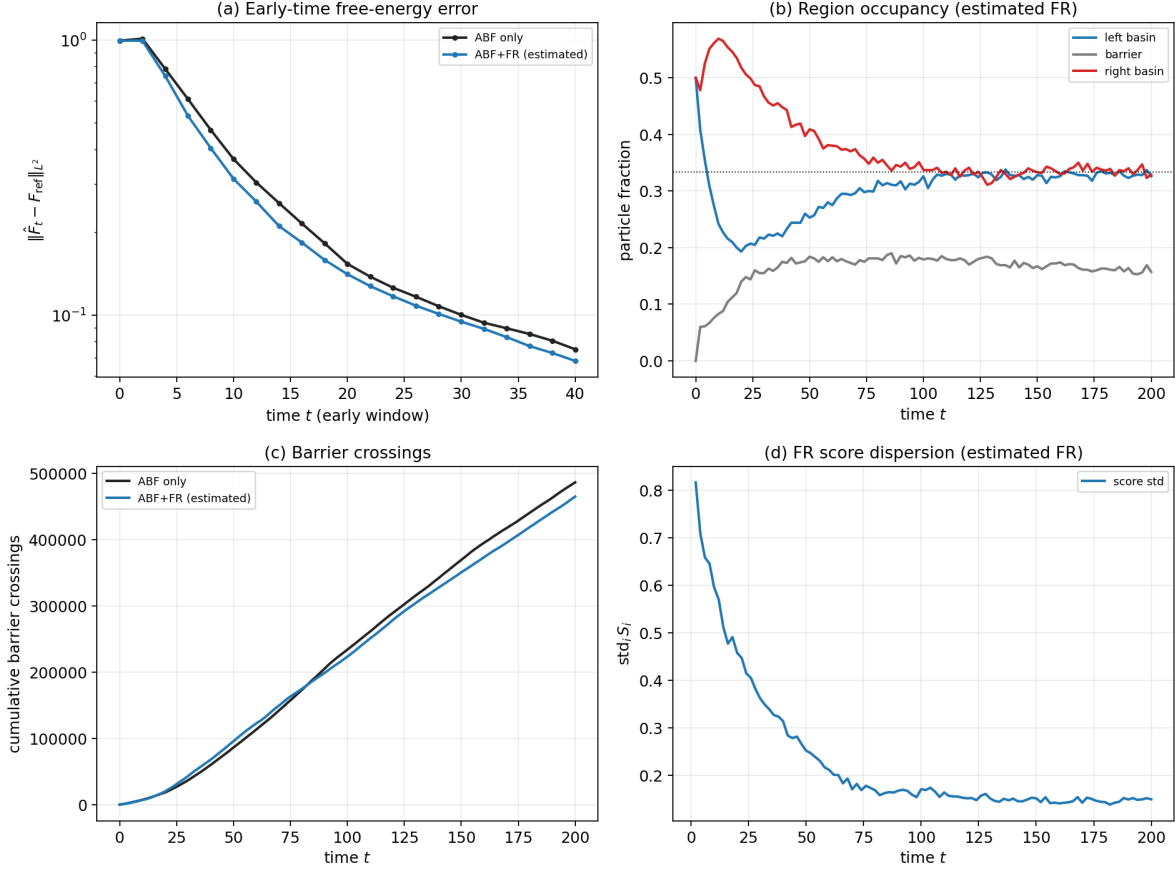


Figure 5: Mechanistic / early-time behaviour, estimated FR (best integrated-rule config) versus ABF only. (a) Early-time free-energy error (log scale): FR converges faster. (b) Region occupancy under FR relaxes from an unbalanced start toward the balanced 1/3 split (dotted). (c) Cumulative barrier crossings are nearly identical — the benefit here is redistribution, not a large increase in raw crossings. (d) FR score dispersion decays as $\hat{p}_t \rightarrow q_t$.

setting does *not* minimise the final error (Section 4.3.2): there is an explicit speed-versus-accuracy trade-off in γ . The marginal bandwidth η is robust at intermediate values, and an early burn-in helps the integrated error here because the largest imbalance is at early times. Across the successful configurations the event fractions are tiny ($\sim 10^{-5}$) and the cap f_{max} never binds, so particle diversity is preserved.

4.5 Case finding

On this easy metastability example the answer to Q1 is *yes, but modestly, and chiefly as a convergence accelerator*. Practical estimated-target FR improves the integrated free-energy error by 11.5% with a smaller final-budget gain; the improvement is real but not a rescue, and it is not unconditional. Because plain ABF already converges well here, the room for FR to help is small. The oracle control does best, which — read *within this regime* — points to target estimation as the practical bottleneck; we will see in Sections 5 and 6 that this reading does not survive into the starved regimes, where the oracle’s advantage vanishes or reverses.

Conditional sampling fidelity at T (aggregated over configs and seeds; not config-resolved)

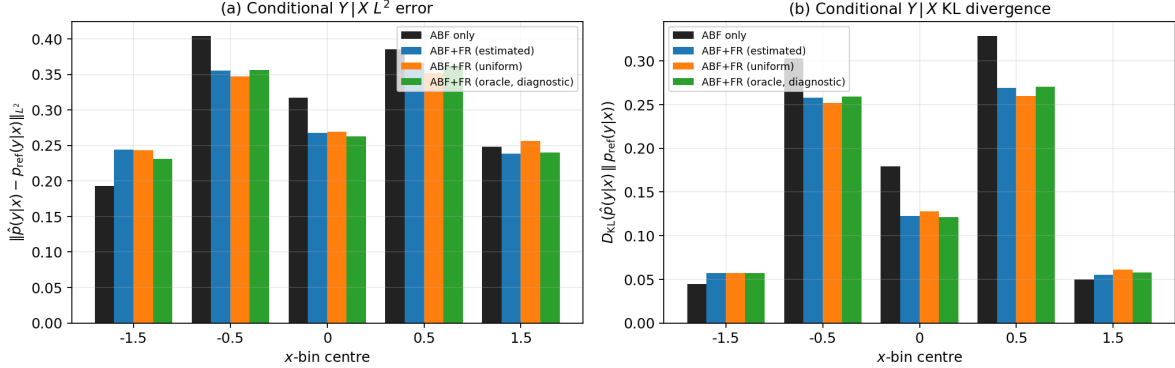


Figure 6: Conditional $Y | X = x$ fidelity at T , by x -bin: (a) L^2 distance and (b) KL divergence of $\hat{p}(y | x)$ to $p_{\text{ref}}(y | x)$. FR does not inflate the conditional error; near the barrier it slightly reduces it. *Limitation:* the conditional CSV identifies runs only by (method, target, seed) and is aggregated over configurations (Remark 3).

5 Case II: an entropic bottleneck

The second system tightens a transverse channel at the barrier top and, by lowering the temperature, drives ABF into a genuinely sample-starved regime. It also has a rare luxury: the conditional law $Y | X = x$ is *analytic*, so we can test Proposition 1 directly rather than against a histogram.

5.1 Model potential and reference

The potential couples a symmetric double well in x to a transverse harmonic channel whose stiffness varies with x ,

$$V(x, y) = H(x^2 - 1)^2 + \frac{1}{2} \omega(x)^2 y^2, \quad \omega(x) = \omega_{\text{out}} + (\omega_{\text{in}} - \omega_{\text{out}}) e^{-x^2/2s^2}, \quad (17)$$

with reaction coordinate $\xi(x, y) = x$. The stiffness $\omega(x)$ is large near $x = 0$ (a *narrow channel at the barrier top*) and relaxes to ω_{out} in the wells. Because the y -integral is Gaussian, the reference is analytic up to a constant:

$$F_{\text{ref}}(x) = H(x^2 - 1)^2 + \beta^{-1} \log \omega(x) + C, \quad F'_{\text{ref}}(x) = 4Hx(x^2 - 1) + \beta^{-1} \omega'(x)/\omega(x), \quad (18)$$

and, crucially, the conditional law is the exact Gaussian

$$Y | X = x \sim \mathcal{N}(0, 1/(\beta \omega(x)^2)). \quad (19)$$

The base regime is $\beta = 8$, $H = 2.5$, $\omega_{\text{out}} = 1$, $\omega_{\text{in}} = 25$, $s = 0.25$, with $N = 256$ walkers, $\text{dt} = 10^{-3}$, and 40 000 steps. The free energy is gauged at $x = 0$ and all L^2 errors are RMS over the interior window $x \in [-1.5, 1.5]$.

Why this is an entropic bottleneck — and an honest caveat. Near $x = 0$ the accessible y -width shrinks like $1/\omega(x)$: at the base setting the transverse channel is $\omega_{\text{in}}/\omega_{\text{out}} = 25$, i.e. its standard deviation is $25\times$ narrower than in the wells, contributing an *entropic* term $\beta^{-1} \log \omega(x)$ to the free energy. We are explicit, however, that in this parameterisation the entropic barrier

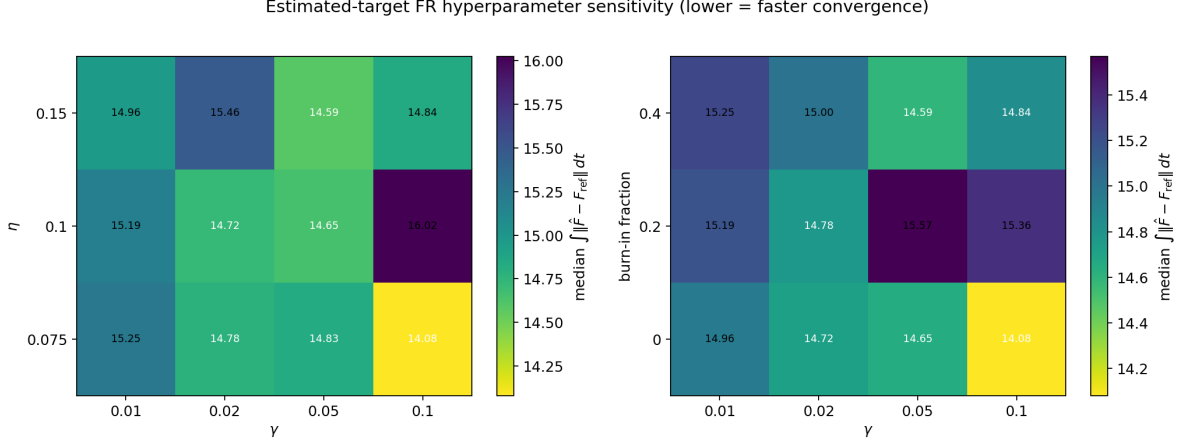


Figure 7: Estimated-target FR hyperparameter sensitivity: median integrated $L^2(F)$ over the grid (lower = faster convergence). Left: γ versus η ; right: γ versus burn-in fraction. The integrated error improves with larger γ and earlier burn-in; intermediate η is robust. These trends should not be extrapolated beyond this regime.

Table 5: Entropic-bottleneck base regime and Fisher–Rao hyperparameters.

Quantity	Value
Inverse temperature β	8
Barrier height H	2.5
Channel stiffness $\omega_{\text{out}}, \omega_{\text{in}}$	1, 25
Channel width s	0.25
Walkers N	256
Time step Δt	10^{-3}
Steps per run	40,000
FR rate γ	15
FR stride <code>fr_every</code>	10
Score clip S_{max}	3
Event cap f_{max}	0.08
Target EMA rate	0.005

$\beta^{-1} \log \omega_{\text{in}} \approx 0.4$ is small next to the energetic barrier $4H \approx 10$ (in F -units). The bottleneck modulates the transverse width but is *not* the dominant contributor to reaction-coordinate starvation; that role belongs to the energetic double well. This matters for reading the ω_{in} sweep below.

5.2 Methods and design

The practical comparison is again **ABF only** versus **ABF+FR estimated** (12), on matched seeds (identical initial conditions and Langevin noise per seed). The diagnostics **FR uniform** (13) and **FR oracle** (14) (built from the analytic F_{ref} , hence not deployable, Remark 2) are also run. The FR hyperparameters are listed in Table 5; an assertion guards every batched call so the reference never reaches the estimated target.

Entropic bottleneck: convergence (median and IQR over seeds)

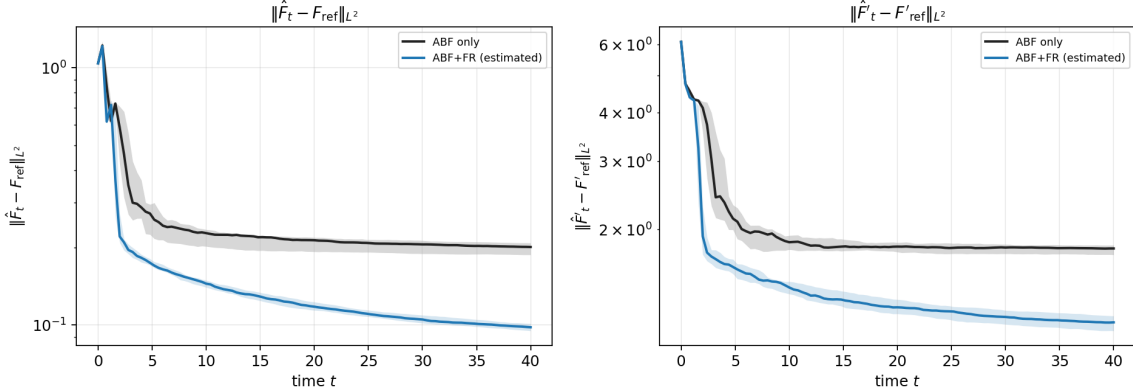


Figure 8: Entropic bottleneck, convergence (median and inter-quartile range over seeds). ABF+FR estimated separates from ABF early and stays below it for the entire trajectory on both the free-energy (left) and mean-force (right) errors.

Table 6: Temperature sweep at $\omega_{\text{in}} = 25$: the FR gain flips sign as ABF crosses from well-sampled (hot) to starved (cold).

β	ABF $\ F - F_{\text{ref}}\ $	FR $\ F - F_{\text{ref}}\ $	gain (%)	win rate
2	0.055	0.060	-8.8	0.30
4	0.049	0.062	-26.1	0.00
8	0.201	0.098	+51.3	1.00
12	0.416	0.281	+32.3	1.00

5.3 Results

At the base regime the practical method reduces the median final free-energy error from 0.201 (ABF) to 0.098 (ABF+FR estimated), a 51.3% reduction, winning 20 of 20 matched seeds (Figure 8). Two diagnostics are immediately revealing. The *uniform* target — which carries no information about the free-energy shape at all — recovers almost the entire gain (50.5%), while the *oracle* target (the *analytic* F_{ref}) is *worse than plain ABF* (−28.1%). This is the opposite of the metastability ranking (Section 4.5): here a target that over-commits to a fixed shape fights the ABF bias instead of merely equalising coverage. The signature points to *balanced resampling* / *variance reduction across the reaction coordinate*, not free-energy-shape steering, as the source of the gain.

5.3.1 Temperature governs the gain, not the bottleneck width

Varying the inverse temperature β at fixed $\omega_{\text{in}} = 25$ (Table 6) is the cleanest mechanistic signal in the study. FR *helps when sampling is hard and hurts when it is easy*: at high temperature ($\beta = 2, 4$) crossings are frequent, ABF already samples the whole coordinate (its error is ~ 0.05), and FR’s birth–death merely injects resampling noise — it is mildly harmful and loses most seeds (−26.1% at $\beta = 4$). At low temperature ($\beta = 8, 12$) crossings are rare, ABF is starved on the far side of the barrier (error 0.2–0.4), and FR’s balanced resampling substantially repairs the coverage, winning every seed. The cross-over sits between $\beta = 4$ and $\beta = 8$.

By contrast, sweeping the bottleneck width ω_{in} at fixed $\beta = 8$ (Figure 9) leaves the gain

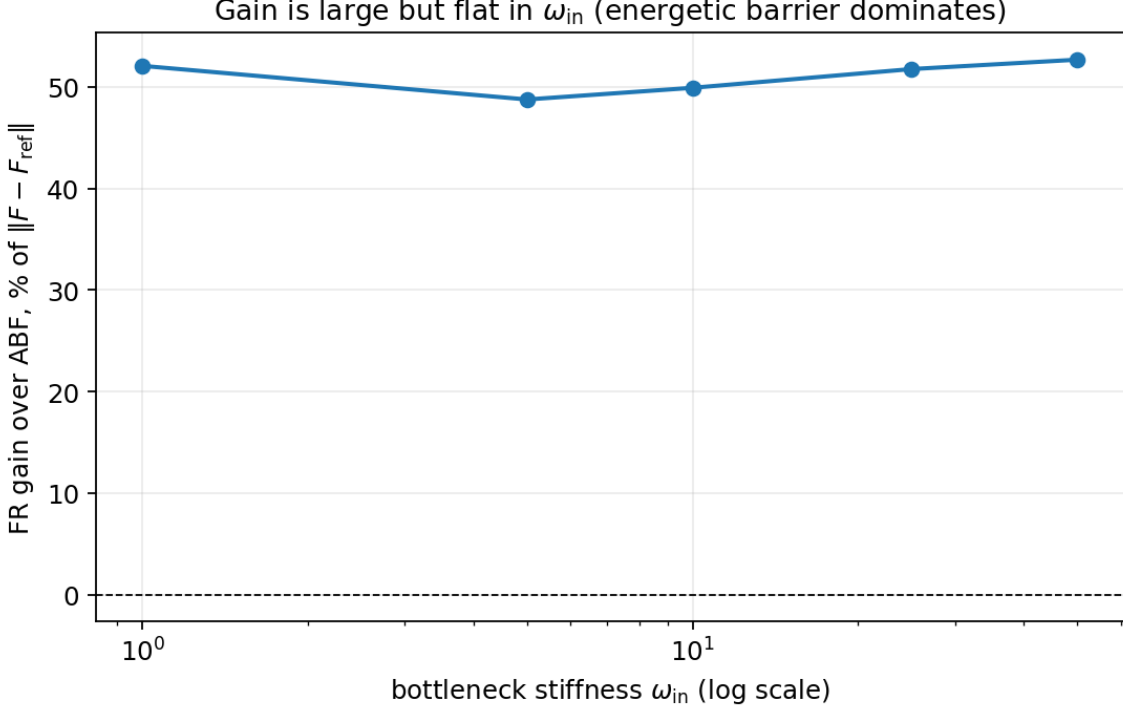


Figure 9: Bottleneck-width sweep ω_{in} at $\beta = 8$. The FR gain is large ($\sim 50\%$) and 100% reliable at every width, but it does *not* grow with the bottleneck strength: the ABF baseline error is flat because the energetic barrier $4H \approx 10$ dominates the sub-unit entropic barrier $\beta^{-1} \log \omega_{\text{in}}$.

essentially *flat* at $\sim 50\%$. This is the honest consequence of the caveat above: the ABF baseline error is nearly independent of ω_{in} because the ω -independent energetic barrier dominates the sampling difficulty. The gain is large and reliable across ω_{in} , but it is governed by metastability (temperature), not by the bottleneck width. We do not claim a bottleneck-strength scaling the data do not show.

5.3.2 A wide safe regime for the FR rate

Sweeping the birth–death rate γ at the base regime (Figure 10), the gain increases *monotonically* up to $\gamma = 50$ (+72.6%); we did not reach a failure boundary within the swept range. The windowed ancestor effective sample size falls steadily from 101 to 9 as lineage diversity collapses, yet the free-energy error keeps improving. This differs from the WCA dimer (Section 6), where aggressive birth–death eventually degrades the estimator; the likely reason is the very short conditional-relaxation time of the analytic y -channel here, so a cloned walker re-equilibrates its orthogonal coordinate almost immediately. FR is unusually robust in this model.

5.4 Mechanism: conditional fidelity from an analytic law

Because $Y \mid X = x \sim \mathcal{N}(0, 1/(\beta\omega(x)^2))$ is *analytic* (19), we can test Proposition 1 directly rather than against a noisy histogram — the gap that Remark 3 left open in the metastability case. Figure 11 compares the empirical $\widehat{\text{Var}}(Y \mid X \in I_j)$ to $1/(\beta\omega(x_j)^2)$ at five x -bins. **FR preserves the conditional law:** at every bin the FR variance tracks the analytic value as well as ABF does, across nearly three orders of magnitude (from $\sim 2 \times 10^{-4}$ at the bottleneck $x = 0$ to ~ 0.12 in the wells), including the extremely narrow channel at $x = 0$. The residual errors are dominated by

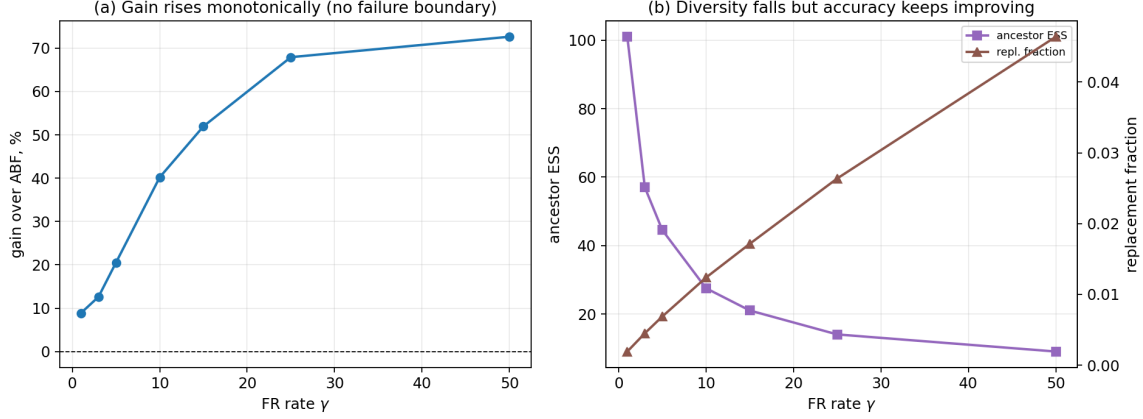


Figure 10: FR-rate sweep $\gamma \in \{1, \dots, 50\}$ at the base regime. Left: gain over ABF increases monotonically (no failure boundary in range). Right: windowed ancestor ESS collapses ($101 \rightarrow 9$) and the replacement fraction rises, yet accuracy keeps improving.

bin-count noise, not by any systematic FR distortion. The free-energy improvement is therefore achieved *without* corrupting the orthogonal degrees of freedom — exactly the benign behaviour Proposition 1 requires, now confirmed against an exact conditional.

5.5 Case finding

The entropic bottleneck is a controlled bridge between the easy metastability case and the hard WCA dimer. Estimated-target FR robustly and substantially reduces the free-energy error in the cold, sample-starved regime ($\sim 51.3\%$, 20/20 seeds), but the gain is governed by *temperature* (FR helps when ABF is starved, mildly hurts when it is not), not by the bottleneck width. The mechanism is balanced resampling / variance reduction, not shape steering: the uniform target nearly matches the estimated one and the oracle target is *worse* than ABF — the reverse of the metastability ranking. The safe FR regime is wide (no failure boundary up to $\gamma = 50$), and, thanks to the analytic conditional, we confirm directly that successful FR does not corrupt $Y \mid X = x$.

6 Case III: a condensed-phase WCA dimer

The third system is many-body and is where ABF is most starved: a dimer that must push a dense solvent aside to change its bond length. It tests whether the two-dimensional findings survive in a setting with thousands of degrees of freedom and a nonlinear collective variable.

6.1 Model and reference

A dimer of two tagged particles sits in a periodic two-dimensional bath of $n_{\text{dim}}^2 - 2$ Weeks–Chandler–Andersen (WCA) solvent particles ($n_{\text{dim}} = 10$, lattice spacing a , box $L = n_{\text{dim}} a$). All particles interact through the purely repulsive WCA potential [10] (Lennard–Jones cut and shifted at its minimum $r_0 = 2^{1/6}\sigma$). The two dimer particles additionally feel a symmetric double-well bond $V_{\text{dim}}(r) = h(1 - ((r - r_0 - w)/w)^2)^2$ with barrier height $h = 2$ and width $w = 2$. The reaction coordinate is the scaled bond length

$$z(q) = \frac{|q_1 - q_2| - r_0}{2w}, \quad (20)$$

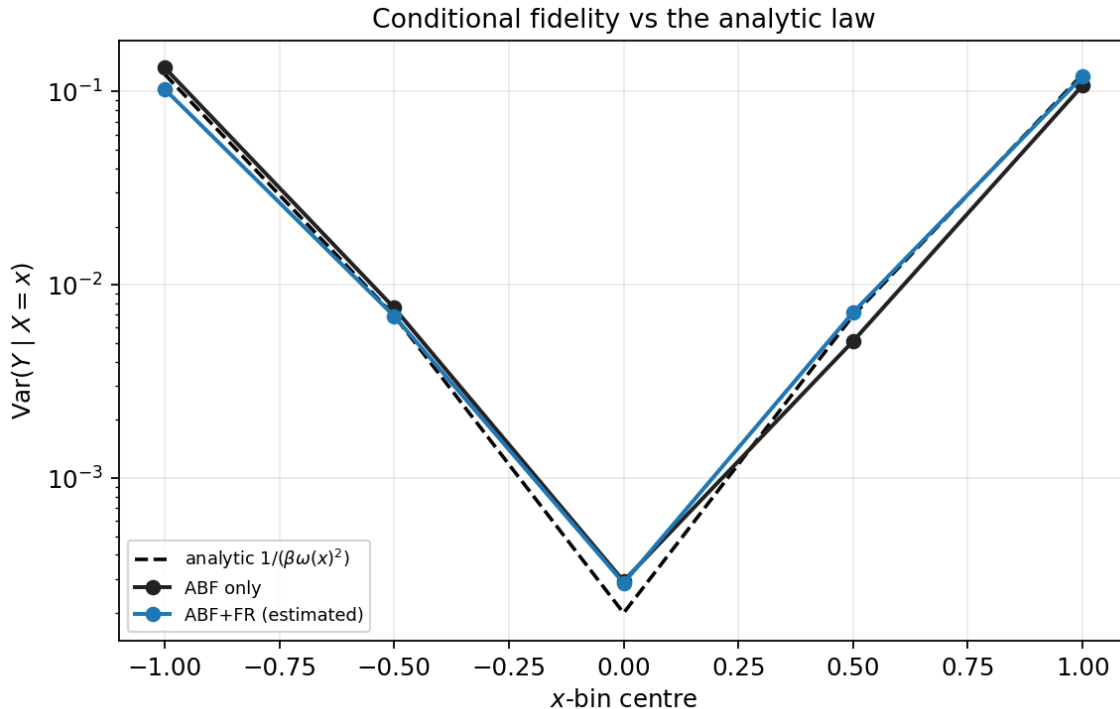


Figure 11: Conditional fidelity against the *analytic* law $\text{Var}(Y | X = x) = 1/(\beta\omega(x)^2)$. Empirical FR and ABF variances both track the analytic curve across three orders of magnitude; FR is not worse than ABF at any bin, including the narrow channel at $x = 0$.

a nonlinear collective variable of the full configuration $q \in \mathbb{R}^{2n}$ (Remark 1): $z \approx 0$ is the *compact* state, $z \approx 1$ the *stretched* state, and $z \in [0.25, 0.75]$ the *transition* region. The free-energy barrier near $z = 0.5$ is partly entropic / crowding-driven, since the dimer must displace solvent to cross it. Dynamics are overdamped Langevin at $\beta = 1$, $dt = 0.002$, with the minimum-image convention; the default system is $a = 1.5$, $\sigma = \epsilon = 1$. The local mean force along z is the constrained projection of the physical forces plus the geometric $2w/(\beta r)$ term (Remark 1). The reference $F(z)$ is computed once per system a by constrained-MD thermodynamic integration [2, 5] and used for evaluation only; errors are taken on the interior window $z \in [-0.1, 1.1]$ with the additive constant aligned.

6.2 Methods and design

As before the practical comparison is **ABF only** versus **ABF+FR estimated** (12), with **FR uniform** and **FR oracle** diagnostics; here the oracle target is built from the TI reference and is explicitly *not* deployable (Remark 2). A no-leakage assertion ensures only the oracle method ever receives the reference, and a clone copies the *entire* WCA configuration — not just z — so cloned solvent arrangements travel with the bond length. Replica lineage is tracked to report the ancestor effective sample size $\text{ESS} = 1/\sum_a w_a^2$. The tuned FR configuration is given in Table 7; the rate is named `fr_rate` and the onset `fr_start_steps` (Section 3.4).

6.3 Results

The main comparison (Table 8; 10 seeds, $N = 1024$, 250 000 steps) shows the practical method **reliably beating ABF**: the tuned FR cuts the median final $L^2(F)$ from 0.0835 to 0.0361 (+55.0%),

Table 7: WCA dimer base system and tuned Fisher–Rao hyperparameters.

Quantity	Value
Inverse temperature β	1
Time step Δt	0.002
Steps per run	250,000
Replicas N	1024
Lattice spacing a	1.5
Dimer barrier h , width w	2, 2
FR rate <code>fr_rate</code>	0.10
FR onset <code>fr_start_steps</code>	20,000
FR stride <code>fr_every</code>	5
Score clip $S_{\max}$	2.0
Event cap $f_{\max}$	0.02
Target EMA rate	0.005

Table 8: WCA main comparison (250k steps, $N = 1024$, $a = 1.5$). The headline gain is in the free energy; the mean-force gain is modest.

Method	$\ F - F_{\text{ref}}\ $	$\ F' - F'_{\text{ref}}\ $	gain (%)	wins	anc. ESS
ABF only	0.0835	0.3220	–	–	–
FR estimated (tuned)	0.0361	0.2513	+55.0	10/10	86
FR uniform	0.0372	0.2525	+54.6	10/10	90
FR oracle*	0.0387	0.2475	+52.7	10/10	100
FR estimated (strong)	0.0329	0.2590	+58.9	10/10	45
FR estimated (aggressive)	0.0935	0.3883	-13.1	1/10	22

* The oracle target uses the thermodynamic-integration reference F_{ref} ; it is a diagnostic control, *not* a deployable method. Medians over 10 seeds.

winning all 10/10 matched seeds. As in the entropic-bottleneck case, the uniform target (+54.6%) and the oracle target (+52.7%) are *essentially as good as* the estimated target — the online estimated target already extracts nearly all the available benefit, and there is no headroom to the oracle. We are explicit that the headline gain is in the *free energy*: the mean-force error $L^2(F')$ improves only modestly ($\approx 22\%$), so we do not claim a large mean-force improvement.

6.4 Mechanism: resampling-driven variance reduction

The diagnostics support the same mechanism as the entropic bottleneck rather than a coverage-reshaping one. Final reaction-coordinate region fractions and the fraction of effective ABF samples in the transition window are nearly identical across ABF, FR estimated, FR uniform and FR oracle: FR does *not* converge to a different stationary z -distribution. Instead the gain appears immediately after FR activates and is sustained for the rest of the run. Figure 14 shows the interpretation: birth–death acts as a *variance-reduction resampler for the ABF estimator*. Replicas stuck in over-represented configurations are replaced by whole-configuration clones of under-represented ones, raising the number of effectively-independent force samples per z -bin even when the marginal looks unchanged. Because the *entire* WCA configuration is copied (Remark 1), clones inject decorrelated solvent arrangements, accelerating convergence of $\hat{F}$ without requiring a different stationary coverage — consistent with uniform $\approx$ estimated $\approx$ oracle.

WCA dimer: convergence (median and IQR over seeds)

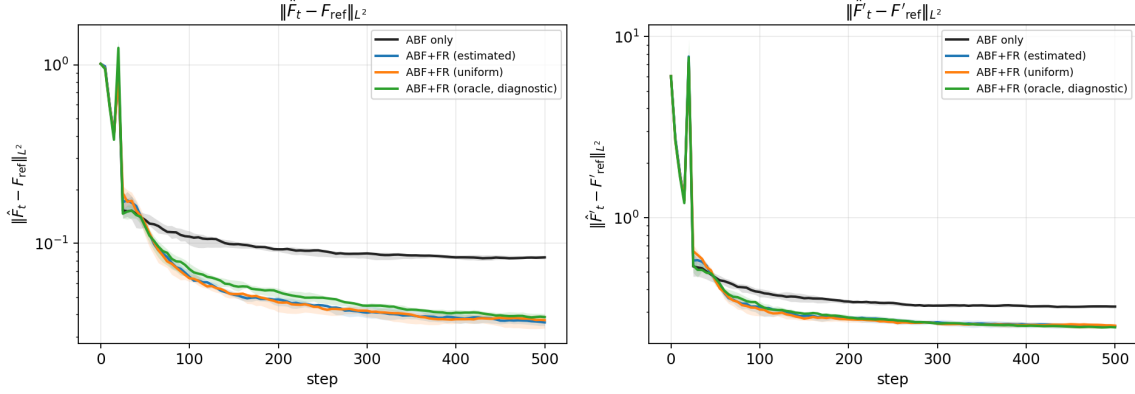


Figure 12: WCA dimer, convergence (median and inter-quartile range over seeds). ABF and FR are identical until FR activates; from then on FR is consistently below ABF on $L^2(F)$ for the rest of the run. The cumulative replacement count (right) confirms the birth–death is active but bounded.

WCA final profiles (median over seeds) vs TI reference

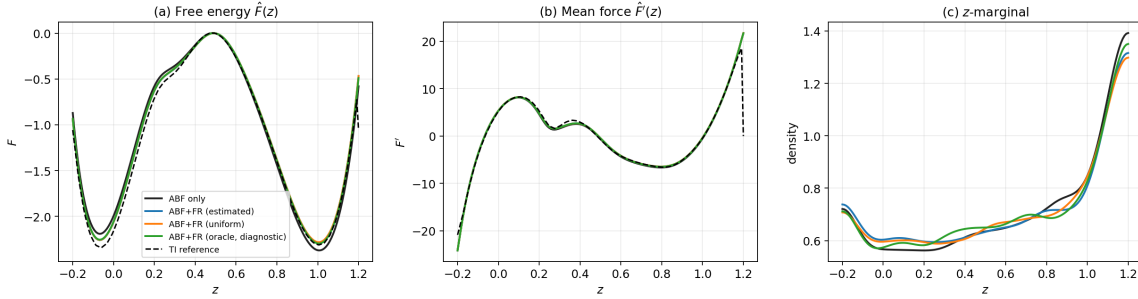


Figure 13: Final profiles (median over seeds) versus the TI reference (dashed): (a) free energy $\hat{F}(z)$, (b) mean force $\hat{F}'(z)$, (c) reaction-coordinate marginal. FR recovers the reference free energy more accurately than ABF, chiefly in and around the transition region.

6.5 Failure boundary

Unlike the entropic bottleneck, the WCA dimer exhibits a clear failure boundary in the FR rate (Figure 15). The rate ladder (a shorter 150k-step, 4-seed stage) maps a clean safe regime: too gentle ($\text{fr_rate} \leq 0.05$) gives a small gain (+19–29%); the sweet spot ($\text{fr_rate} \approx 0.1\text{--}0.2$) gives the maximal +58–64% with healthy ancestor ESS; the gain then *erodes* sharply ($\text{fr_rate} = 0.5$ falls to +12% with the error already climbing back toward the ABF baseline) and finally *reverses* for $\text{fr_rate} \geq 1$, where the median final $L^2(F)$ is worse than ABF (−38% at rate 1, −68% at rate 2) and no seed wins, as the ancestor ESS collapses from ~ 337 to ~ 20 . The reversal arrives even earlier under a longer budget: the production aggressive configuration ($\text{fr_rate} = 0.5$ at 250k steps, 10 seeds) is already net-harmful (−13.1%, winning only 1/10 seeds, ancestor ESS ~ 22), because the extra steps give diversity loss more time to dominate. Over-aggressive birth–death destroys the replica diversity the mean-force estimator depends on, so the estimator degrades faster than coverage improves. This is the predicted failure mode; the analytic y -channel of the entropic bottleneck masks it (Section 5) because cloned walkers re-equilibrate instantly, whereas a dense

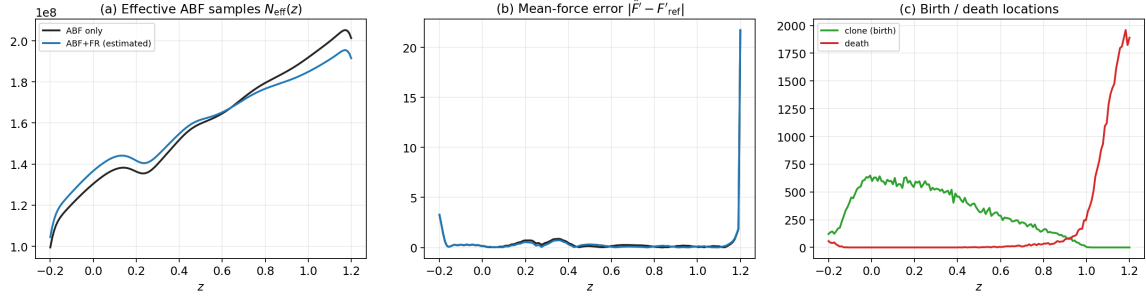


Figure 14: Mechanism. (a) Effective ABF sample count $N_{\text{eff}}(z)$ is raised by FR in the transition window; (b) per-bin mean-force error is reduced there; (c) birth (clone) and death locations concentrate around the over/under-sampled regions; FR reshapes the *sampling quality*, not the stationary marginal.

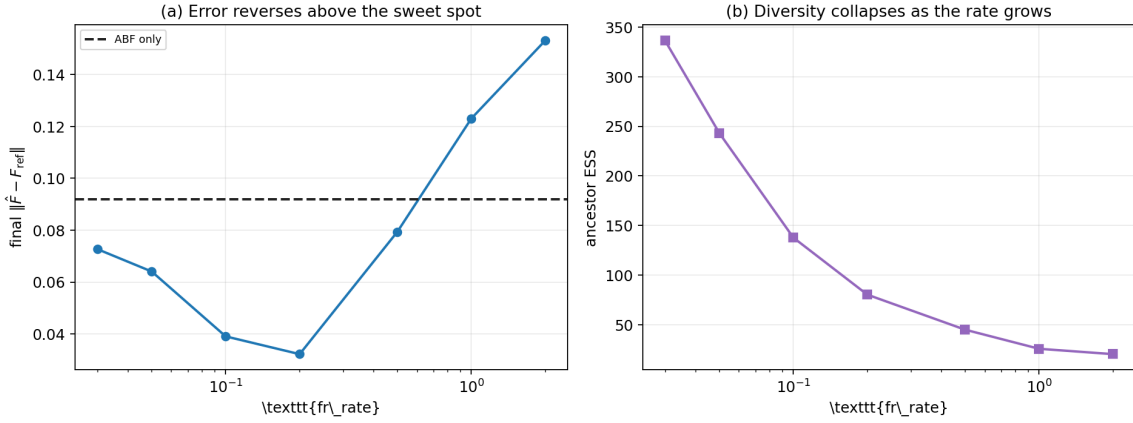


Figure 15: FR-rate ladder (150k steps, 4 seeds). Final $L^2(F)$ (left) is minimised in the 0.1–0.2 sweet spot, climbs back toward the ABF baseline (dashed) by $\text{fr_rate} = 0.5$, and rises above it for $\text{fr_rate} \geq 1$; ancestor ESS (right) collapses as the rate grows, pinpointing diversity loss as the failure mechanism.

solvent does not.

6.6 Difficulty dependence

Varying one difficulty axis at a time (Figure 16) shows the gain is robust and, importantly, *grows with difficulty*. With *budget*, FR wins at every n_{steps} and the relative advantage grows rather than washes out (+35.5% at 50k to +56.9% at 400k): ABF’s error plateaus around 0.08 after ~ 150 k while FR keeps descending, so FR is not merely an early-time accelerator that ABF later catches. With *replica count*, the gain is roughly flat (+51–59% from $N = 256$ to 2048) — FR helps even at $N = 256$. With *solvent crowding*, the densest bath ($a = 1.35$) makes the entropic barrier severe and ABF $\sim 4\times$ worse in absolute error; FR still wins 5/5 and delivers by far the largest absolute error reduction ($0.365 \rightarrow 0.225$). In the units a practitioner cares about — absolute free-energy error removed — the crowded regime is where FR helps most.

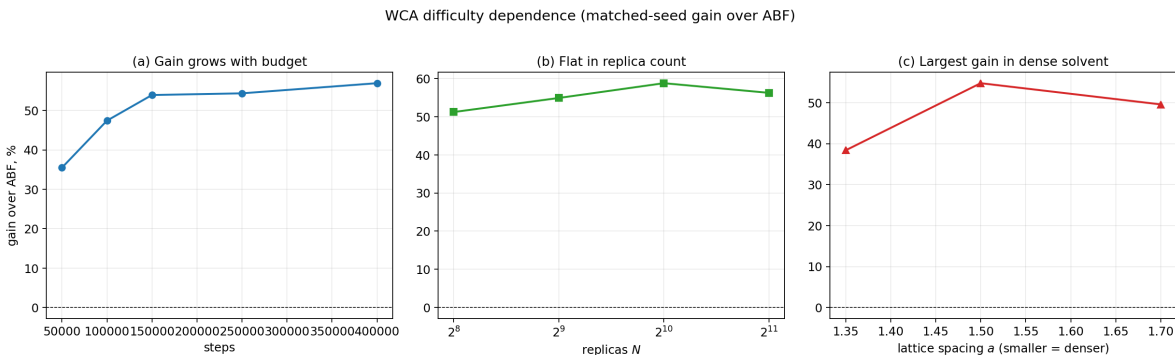


Figure 16: Difficulty dependence (matched-seed gain over ABF). (a) Budget: the relative gain grows with n_{steps} as ABF plateaus and FR keeps improving. (b) Replica count: gain is roughly flat. (c) Solvent crowding: smaller a (denser) yields the largest absolute error reduction. FR wins 5/5 seeds in every cell.

6.7 Case finding

On the condensed-phase WCA dimer, ABF+FR with an online estimated target reliably and substantially improves free-energy accuracy: +55.0% in median final $L^2(F)$ at the tuned operating point, winning 10/10 seeds, with the gain concentrated in the free energy (not the mean force). The mechanism is balanced-resampling variance reduction after warm-up; the precise target shape matters little (uniform $\approx$ estimated $\approx$ oracle, and the estimated target already reaches the oracle’s accuracy). The method has a clear safe regime, `fr_rate` $\approx$ 0.1–0.2, and a real failure boundary: too-aggressive birth–death reverses the benefit as diversity collapses. The advantage *grows with simulation budget*, is insensitive to replica count, and is largest in the densest, hardest solvent — exactly the starved regime the variance-reduction mechanism predicts.

7 Cross-case synthesis

The three cases were chosen to span a single axis: *how sample-starved ABF is*. Read together they tell a more coherent story than any one of them does alone, and they resolve an apparent contradiction in how the FR *target* behaves.

7.1 A starvation spectrum

Order the cases by how badly plain ABF is starved of reaction-coordinate samples. At the easy end is the metastability model (Section 4), where ABF already converges well; in the middle is the entropic bottleneck (Section 5), whose difficulty we can dial with temperature; at the hard end is the condensed-phase WCA dimer (Section 6), where the dimer must displace a dense solvent to cross the barrier. Table 9 lays the headline numbers side by side.

7.2 The gain grows with starvation

The practical, estimated-target FR gain tracks the starvation axis almost monotonically. Where ABF is a strong baseline the gain is small and not unconditional — +11.5% on the integrated error in the metastability case, with a marginal final-budget gain — and where ABF is even less stressed (the hot entropic bottleneck, $\beta \leq 4$) FR is mildly *harmful* (−26.1% at $\beta = 4$), because its

Table 9: Cross-case synthesis: the estimated-target FR gain grows with how sample-starved ABF is, while the value of the target *shape* (oracle vs. estimated) shrinks. Metastability gain is on the integrated error; the others on final $\|F - F_{\text{ref}}\|$.

Case	Regime	FR gain (%)	wins	target effect
Metastability	ABF strong	+11.5 (int.)	–	oracle best
Entropic bottleneck (hot, $\beta=4$)	ABF easy	-26.1	0/10	n/a
Entropic bottleneck (cold, $\beta=8$)	ABF starved	+51.3	20/20	oracle worse
WCA dimer	ABF starved	+55.0	10/10	oracle $\approx$ est.

birth–death injects resampling noise into an estimator that was not starved to begin with. Where ABF is starved the gain is large and reliable: +51.3% in the cold bottleneck (20/20 seeds) and +55.0% on the WCA dimer (10/10 seeds), the relative gain growing further with simulation budget and the largest *absolute* error removed in the densest solvent. The entropic-bottleneck temperature sweep is the cleanest evidence because it moves along the axis *within a single system*: the same model, the same code, the gain flipping sign from $\beta = 4$ to $\beta = 8$ as ABF crosses from well-sampled to starved.

7.3 Why the target shape matters less than it first seemed

The cases appear to disagree about the FR *target*. In the easy metastability case the oracle target is best and the estimated–oracle gap reads as a “target-quality bottleneck” (Section 4.5). In the WCA dimer the oracle is no better than the estimated or uniform target, and in the entropic bottleneck the oracle is actually *worse than plain ABF*. These are not contradictory once the mechanism is separated from the target.

The *dominant* effect of FR is balanced-resampling variance reduction: killing over-represented and cloning under-represented whole configurations raises the number of effectively independent force samples per reaction-coordinate bin, which is exactly what a starved ABF estimator lacks. This effect needs only a roughly correct *direction* of imbalance, not an accurate free-energy shape — which is why the uniform target nearly matches the estimated one in both starved cases. Target-shape steering is a *secondary* effect that becomes visible only when the variance-reduction headroom is already exhausted, i.e. when ABF is well-sampled: there the residual error is limited by the target *direction*, so the oracle (which knows the exact direction) helps. When ABF is starved the residual error is variance-limited, the exact shape is irrelevant, and a target that over-commits to a fixed shape — the oracle, anchored to a F_{ref} that the running bias has not yet matched — can actively fight the bias and lose to a neutral uniform target. The metastability “target-quality bottleneck” is therefore a regime-specific reading, not a general law; we are careful not to carry it into the starved regimes.

7.4 A single failure mechanism: diversity collapse

Both the safe regimes and the one observed failure are explained by the same quantity — ancestor diversity. Aggressive birth–death erodes the effective number of independent lineages; the gain survives only while the orthogonal degrees of freedom re-equilibrate fast enough to re-decorrelate the clones. The WCA dimer reaches the failure boundary ($\text{fr_rate} \geq 1$ reverses the benefit in the rate ladder, and even $\text{fr_rate} = 0.5$ becomes harmful at production budget, ancestor ESS $\rightarrow 20\text{--}22$) because its dense solvent re-equilibrates slowly: clones stay correlated and the estimator degrades. The entropic bottleneck does *not* reach a failure boundary even at $\gamma = 50$ (ESS $\rightarrow 9$) because

its analytic y -channel re-equilibrates almost instantly, so even heavy cloning does not starve the estimator within the budget. Same mechanism, different orthogonal relaxation times — which is why a safe FR rate must be calibrated to how quickly the orthogonal coordinates of *the system at hand* mix.

8 Discussion, limitations, and future directions

8.1 What the three cases establish (Q4)

The original single-case study could only ask whether FR was “icing on the cake” (a mild accelerator when ABF already works) or “snow in the winter” (a rescue when ABF alone struggles), and explicitly deferred the question to harder regimes. The two new cases *are* those harder regimes — lower temperature, fewer particles, and a condensed phase — and the answer is now *both*: FR is icing for easy ABF (metastability; hot bottleneck) and snow for starved ABF (cold bottleneck; dense WCA dimer), with the transition governed by how sample-starved ABF is (Section 7).

8.2 Limitations

We remain deliberately conservative about scope.

1. **The reaction coordinates are low-dimensional.** All three systems use a one-dimensional reaction coordinate (a Cartesian component in the 2D cases, a scalar bond length in WCA). Multi-dimensional or strongly curved collective variables, where the geometric mean-force term and the conditional sampling are far harder, are untested.
2. **The oracle target is not a method.** In every case the oracle uses the reference free energy, the unknown being computed, and is reported only as a diagnostic control (Remark 2). Its behaviour is read only comparatively; notably, it is *not* uniformly best — it is worse than ABF in the entropic bottleneck — which is itself part of the synthesis, not a deployable result.
3. **Production estimators are validated surrogates.** The 2D GPU backend uses a binned-smoothed mean-force/marginal estimator and a vectorised fixed- N birth–death; the WCA backend uses a constrained projection with the geometric term. These pass their validation suites and agree with the kernel/TI references to within tolerance, but are not bit-identical to the original kernel notebook implementation.
4. **Metastability conditional diagnostics are not configuration-resolved** (Remark 3). The entropic-bottleneck case closes this gap with an analytic conditional law, but only for that model; the WCA conditional fidelity is argued structurally (Remark 1) rather than measured per configuration.
5. **The entropic barrier is sub-dominant in Case II.** As stated in Section 5, the bottleneck modulates the transverse width but the energetic double well dominates the difficulty; the ω_{in} sweep is honestly flat, and we do not claim a bottleneck-strength scaling.

8.3 Failure modes

The analysis points to a single unifying failure mechanism, diversity collapse, plus target misspecification (Section 7). First, an *overly aggressive mechanism* erodes ancestor diversity faster than the orthogonal coordinates can re-decorrelate clones; this is realised in the WCA dimer for $\text{fr_rate} \geq 1$

in the rate ladder (the benefit reverses, ancestor ESS $\rightarrow$ 20), with even `fr_rate`= 0.5 turning harmful at production budget, and is masked in the entropic bottleneck only because its analytic channel re-equilibrates instantly. The safe FR rate must therefore be calibrated to the orthogonal relaxation time of the system. Second, a *noisy or biased target* caps the attainable benefit; the metastability oracle-estimated gap quantifies the headroom in the easy regime, while the entropic-bottleneck oracle shows that in a starved regime an over-committed target can do worse than a neutral uniform one.

8.4 Future directions

- **Adaptive, diversity-aware FR strength.** Tie `fr_rate`/ γ to a measured ancestor-ESS or orthogonal-relaxation signal, so the rate self-limits before diversity collapses — capturing the early-time speed of an aggressive rate without the late-time failure.
- **Higher-dimensional and curved reaction coordinates**, where conditional sampling is genuinely hard and the geometric term is nontrivial.
- **Configuration-resolved conditional diagnostics** for the many-body case, to measure (not just argue) conditional fidelity per hyperparameter.
- **Sharper entropic stressors.** A model in which the entropic barrier dominates the energetic one would test whether the bottleneck width, not just the temperature, can drive the gain.

9 Conclusion

Across three controlled systems with essentially exact references — a 2D metastability model, a 2D entropic bottleneck with an analytic conditional law, and a many-body condensed-phase WCA dimer — a Fisher–Rao birth–death correction helps the Adaptive Biasing Force method *in proportion to how sample-starved ABF is*. When plain ABF already samples well the practical, estimated-target FR is a modest convergence accelerator and can even be mildly harmful (the metastability model, +11.5% integrated error; the hot entropic bottleneck, −26.1%). When ABF is starved the same correction is a large and reliable win: +51.3% in the cold bottleneck (20/20 seeds) and +55.0% on the WCA dimer (10/10 seeds), with the advantage growing with simulation budget and solvent density.

The mechanism is the same in all three cases and is best described as *balanced-resampling variance reduction*: by cloning whole configurations from under-represented regions of the reaction coordinate, FR raises the number of effectively independent force samples per bin, which is exactly what a starved ABF estimator lacks. This explains the study’s most counter-intuitive observation — that the precise FR *target* matters little in the starved regimes, where a uniform target nearly matches the estimated one and an oracle target is no better (WCA) or even worse (entropic bottleneck) than ABF. Target-shape steering is a secondary effect that surfaces only when the variance-reduction headroom is spent, i.e. when ABF is well-sampled; the “target-quality bottleneck” reading of the easy metastability case does not generalise. The single failure mode is diversity collapse under too-aggressive birth–death (the WCA dimer’s gain erodes past the `fr_rate` $\approx$ 0.1–0.2 sweet spot and reverses outright for `fr_rate` $\geq$ 1), bounded by how fast the system’s orthogonal coordinates re-equilibrate cloned replicas.

Two caveats persist throughout. The oracle target uses the reference free energy and is never a deployable method; it is reported only as a diagnostic whose behaviour — best in the easy case, worst in the cold bottleneck — is itself part of the synthesis. And the practical recommendation is

unambiguous: use the estimated target, at a moderate rate calibrated to the orthogonal relaxation time, and expect the largest returns precisely where ABF alone struggles most.

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